06 THE KOI & LOTUS POND

LIVING WATER AT THE THRESHOLD

An almond-shaped pond greeting you before the Nur Dome's door.

LEVEL	CREW	WHERE	STAGE NOW
■■□ Some skill or supervision	2–5	Just before and left of the Nur Dome entrance	Location, size, dig plan

The almond-shaped koi and lotus pond sits just before and to the left of the Nur Dome's entrance. It is the living, planted pond (fish, lilies, lotus) as a contrast to the still mirror of the reflecting pool further in. Bonus: the soil you dig out here feeds the dome and the wall.

FIRST STEPS *today*

1. With Yusuf, lay out the almond with a garden hose. Check sun: lotus needs at least 6 hours of full sun. Avoid spots under trees.
2. Measure the gap to the dome trench (at least 6 ft) and decide where overflow water will go in a big rain.
3. Paint the outline. Mark the shelf line and the deep zone inside it.
4. Dig in stages: the whole area to shelf depth first, then the deep zone. Spoil goes to the dome and wall earth piles.
5. Check that the rim is level all the way round (water level). An unlevel rim shows liner on one side.

BEFORE YOU START

- Call 811 (DigAlert) before digging and confirm where the water lines run.
- Keep the pond edge at least 6 ft from the dome's foundation trench. Saturated ground next to an earthen wall is a real risk to the dome.
- Deep water and children or guests: plan a planting shelf and stones so no one steps straight into deep water.

THE BASICS

Size	Koi grow to 2 ft or more. For 3–5 koi, aim for at least 1,500 gallons and 3 ft deep at the center (4 ft is better: cooler in summer, safer from herons and raccoons).
Example	An almond about 14 × 7 ft, 3½ ft deep at the center, with a 10–12 in deep planting shelf on the sunny side: roughly 1,200–1,600 gallons. Rough gallons = length × width × average depth × 7.5 × 0.6 for an almond shape.
Liner	45-mil fish-safe EPDM over an underlayment (old carpet works). Liner size = length + 2 × depth + 2 ft, by width + 2 × depth + 2 ft. For the example: about 23 × 16 ft.
Natural filter	A bog filter: a gravel bed planted with marginal plants, about 10–20% of the pond's surface area, with pond water pumped up through it. Low-maintenance and traditional. Size the pump to move the whole pond every 1–2 hours (750–1,500 gallons per hour for this size).
Oxygen 24/7	Koi need the pump or an aerator running at night. A solar pump without a battery is fine for fountains, not for koi.

HOW-TO NOTES

Lotus vs. koi	Koi dig up and eat plants. Plant lotus in wide, heavy pots with no holes, top the soil with river rock, and set the pots on the shelf 2–6 in below the surface to start. Lotus is planted from tubers in spring (March–May); order this winter. Water lilies are a hardy companion.
Water	The pond can lose 1–2 in a week to evaporation in summer. Agree a top-up plan with the neighbor who shares the tank.
Fish	Let the pond run 4–6 weeks before adding fish, so beneficial bacteria build up. Start with 3 small koi.

THE PLAN *in order*

01	**Site + shape**	Hose layout, sun check, distance from dome, overflow route.
02	**Dig**	Shelf depth, then deep zone; spoil to the build piles; rim leveled.
03	**Line**	Underlayment, EPDM liner, rocks on the shelf and edge.
04	**Plumbing**	Pump, bottom return, bog filter, skimmer.
05	**Fill + run**	Fill, run two weeks, check for leaks.
06	**Plant**	Marginals in the bog, lilies; lotus in spring.
07	**Fish**	After 4–6 weeks of cycling, 3 small koi.

TOOLS & MATERIALS

Garden hose, marking paint · Shovels, mattock, wheelbarrows · Water level · Underlayment + 45-mil EPDM liner · Pond pump, tubing, bog gravel · River rock and flat stones

FOR YUSUF TO DECIDE

- ? Final size and depth
- ? Bog filter location
- ? How the pump is powered at night
- ? Number of koi
- ? Lotus varieties

Resources: Koi club referrals: Associated Koi Clubs of America (akca.org) · Koi + water gardens: Mystic Koi & Water Gardens, Upland